

Nicole Gerzon

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EDUCATION

Northeastern University, Khoury College of Computer Science , Boston, MA	Sept. 2024 - present
<i>PhD Student in Cybersecurity</i>	<i>GPA: 4.00/4.00</i>
Northeastern University , Boston, MA	Sept. 2019 - May 2024
<i>Bachelor of Science in Cybersecurity</i>	<i>GPA: 3.92/4.00</i>

ACADEMIC RESEARCH

Spotify Recommendations (<i>in progress</i>)	July 2025 - present
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- Investigating Spotify's podcast recommendation algorithm to uncover the underlying structure of recommendations and how user activity affects episode and show recommendations.
- Developing automated, repeatable methods of auditing both the mobile and web app.

Privacy in Connected Vehicles (<i>under submission</i>)	Jan. 2024 - present
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- Developed methodology for measuring network traffic generated by connected vehicles and collected data in collaboration with a corporate partner from 30 different vehicles, spanning 29 different manufacturers.
- Conducted a comprehensive analysis of data generated by connected vehicles, focusing on privacy, security, and data-sharing practices
- Reverse-engineering mobile and in-vehicle apps to determine how personal/vehicle data is collected & processed

Quantifying the Threat of Sandwiching MEV on Jito:

A Measurement of Solana's Leading Validator Client

Jan 2025 - Aug. 2025

Published at AMC IMC 2025

- Reverse engineered hidden APIs to collect 4 months worth of otherwise opaque Jito-Solana transaction data, identifying over 500k Sandwiching MEV attacks, over \$7.7 million in cumulative user losses.
- Found evidence of user defensive activity totaling \$2.4 million and correlated to a decrease in attacks.

Track You: A Deep Dive into Safety Alerts for Apple AirTags

Mar. 2022 - July 2023

Published at PETS 2023

- Created methods to analyze the underlying structure and characteristics of Apple's FindMy network infrastructure
- Developed a process for creating stealthy, counterfeit AirTags and linking them to real keys, demonstrating flaws in Apple's data storage and security protocols
- Conducted BLE analysis to investigate existing behavior of Apple AirTags and study their behavior in different states of the pairing lifecycle

WORK EXPERIENCE

Texas Office of the Attorney General, Austin, TX

Cybersecurity Intern, Child Support Division

Aug. 2023 - Aug.2024

- Facilitated compliance audits for IRS, CJIS, and various state-wide regulatory requirements
- Validated implementation and ensured documentation of NIST, FedRAMP, and TexRAMP security controls and policies across IT organization and third party vendors
- Scanned for and analyzed risk of open vulnerabilities across division infrastructure and state-wide offices for patch management purposes and presenting security information in weekly advisory board meetings
- Led the organization and optimization of the incident response system to improve tracking and management of incoming incidents

Liberty Mutual, Portsmouth, NH

Cybersecurity Intern - Threat Defense and Response Team

June 2022 - Aug 2022

- Aided in machine hardening efforts by auditing usage and structure of unsigned, insecure scripts across all 50,000 company machines using CrowdStrike (PSFalcon/Powershell)
- Proposed and implemented company-wide system rules for restricting unauthorized command line use.
- Automated database auditing through python and SQL scripts